

Theorem 1 (Division with remainder). *Let $a \in \mathbb{Z}$ and $b > 0$. There exist unique q, r with*

$$a = qb + r \quad \text{and} \quad 0 \leq r < b.$$

Proof. Existence. Take the least non-negative element of the set of values $a - kb$; it exists by well-ordering.

Uniqueness. Two such remainders differ by a multiple of b that is smaller than b , hence by zero. $\square$

Corollary 2. *Every integer is either even or odd, and not both.*

1 Greatest common divisors

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